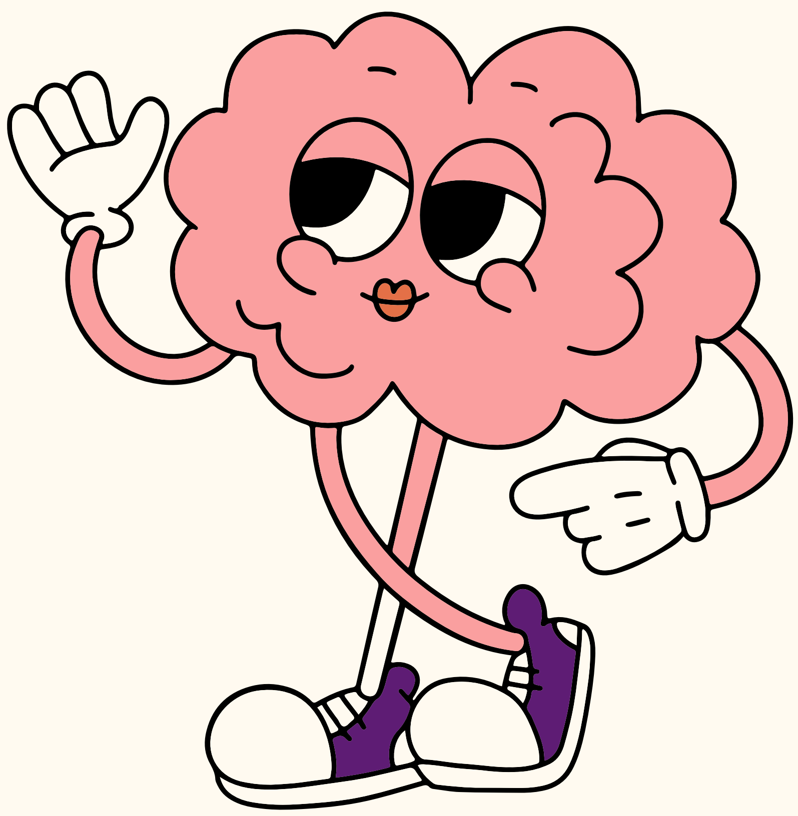
@jahnavivetukuri

DESIGN PATTERNS

in Java

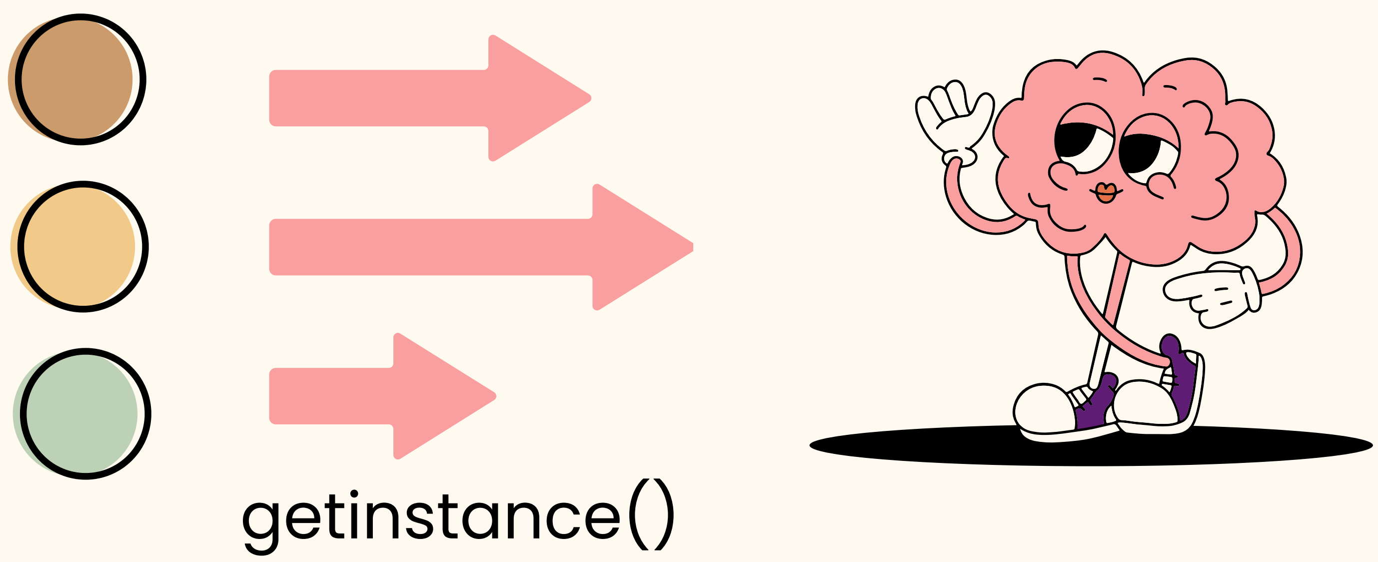
Create more maintainable, scalable, and flexible code



@jahnaviVetukuri

SINGLETON PATTERN

Ensures that a class has only one instance and provides a global point of access to that instance

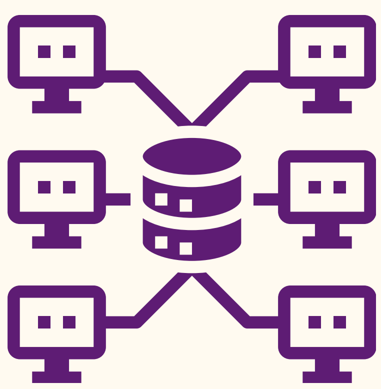


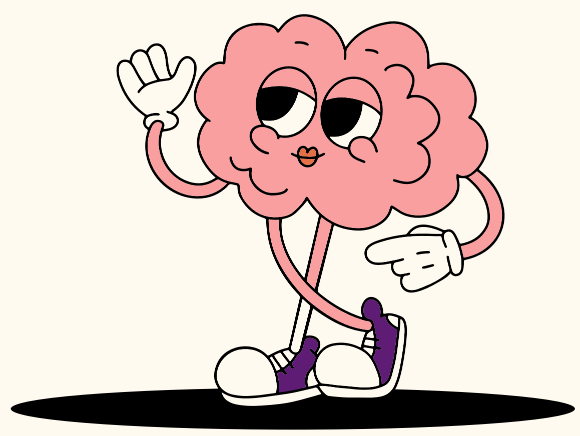
USECASE

DATABASE CONNECTION POOL

In a web application, you need to manage a pool of database connections to optimize resource usage and improve response times.

The Singleton pattern ensures that there's only one central connection pool manager instance.





FACTORY PATTERN

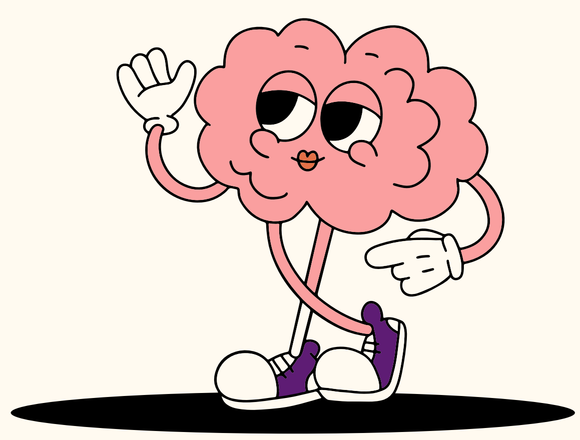
The Factory pattern separates object creation from its use, providing a way to create objects without specifying the exact class.

USECASE

DOCUMENT CONVERSION SERVICE

Consider a document conversion service that converts various document formats (e.g., PDF to Word, Excel to PDF, etc.).

The Factory pattern abstracts the creation of specific document converter classes.



OBSERVER PATTERN

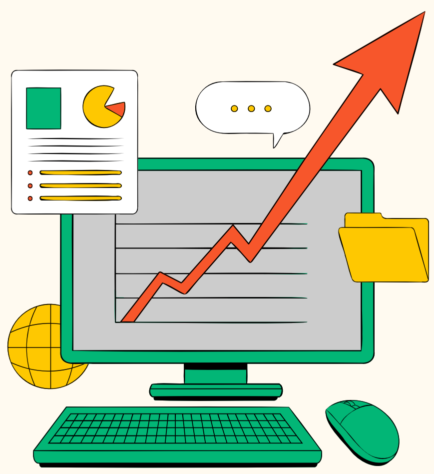
The Observer pattern defines a one-to-many relationship between objects so that when one object changes state, all its dependents are notified and updated automatically.

USECASE

STOCK MARKET UPDATES

In a stock market application, multiple subscribers (e.g., traders or investors) want to receive real-time updates on stock prices.

The Observer pattern is used to implement a notification system where the stock market serves as the subject, and the subscribers are observers.



REPOST IF FOUND USEFUL

FOLLOW FOR MORE

@jahnavivetukuri